

Scientific discovery has always pushed back the frontiers of knowledge and solved pressing human problems, but technological progress also raises serious ethical dilemmas—from the morality of genetic technology, to questions of regulation, such as censorship of the Internet, or the ecological damage generated by industrialization. Scientific inventions have often generated changes undreamt of by their makers. Albert Einstein (see pp.416–17), the brilliant physicist who formulated the Theory of Relativity, saw his discovery used to create the atomic bomb. He later remarked, “It has become appallingly obvious

“Almost everything that **distinguishes the modern world** from earlier centuries is attributable to **science**.”

BERTRAND RUSSELL, BRITISH PHILOSOPHER, “A HISTORY OF WESTERN PHILOSOPHY,” 1945

that our technology has exceeded our humanity.”

Aerospace

Space travel is perhaps the greatest symbol of technological progress in the modern age. The race between the two opposing Cold War (see pp.406–07) superpowers, the US and the Soviet Union, for supremacy in space

gathered pace in the late 1950s and ended when Soviet astronaut Yuri Gagarin successfully orbited Earth in 1961. The next goal was to put a man on the Moon. This was achieved in 1969 when the US Apollo 11 spacecraft landed on the Moon, and millions across the world watched astronauts Neil Armstrong and Buzz Aldrin step onto the lunar surface. In 1975, the superpowers adopted a policy of cooperation and simultaneously launched crewed spacecraft, a mission known as the Apollo–Soyuz Test Project.

Volkswagen Beetle

The “Beetle” was first produced in Germany in 1938 as Adolf Hitler’s affordable “people’s car.” Production of the classic VW Beetle peaked at 1.3 million in 1971 and ceased in 2003. It remains an icon of twentieth-century design.



DISCOVERY

NUCLEAR FISSION

Nuclear fission occurs when the nucleus of an atom splits into two or more smaller nuclei. It is used in the production of nuclear power and weapons. Scientists Otto Hahn and Fritz Strassman discovered the process during research carried out in 1938–39. In 1942, a team led by Enrico Fermi created the first controlled self-sustaining nuclear chain reaction. With World War II raging, governments funded scientists to develop an atomic bomb. In the US, physicist Robert Oppenheimer led the Manhattan Project, which built the bombs used on Japan (see p.405).



Modern Technology

Technological advances continue to transform daily life. Today we can understand the workings of the universe down to its tiniest particles and change the nature of living organisms. We live longer and travel farther and faster than the most optimistic of inventors would have predicted in earlier centuries.

BEFORE

Scientific discoveries have at times caused controversy, but advances in radio, computing, and new materials in the 19th century made future breakthroughs possible.



BAKELITE TELEPHONE

Herman Hollerith used punchcard techniques in the Hollerith Desk, a prototype computer.

SYNTHETIC MATERIALS

Many materials used in modern technology were invented in the 19th century, including parkesine, the first manufactured plastic, in 1855, and cellulose triacetate, the earliest synthetic fiber, in 1869, a forerunner of rayon, nylon, and polyester. In 1907 Leo Baekeland created the first plastic made from synthetic materials and named it Bakelite.

Subsequent collaborations have included the European Space Station, *Mir*, and the International Space Station, which draws on the scientific and technological resources of 16 nations.

New materials

Most modern manufactured items are made from materials specially developed to be durable, portable, and efficient. The production of so-called “smart materials” has had a big impact in recent times. Smart materials have the capacity to alter their properties when external conditions change. They include smart alloys or “shape memory alloys,” metals that “remember” their own geometry and can return to their original shape after being distorted.



First views of the moon

An American boy watches as the first televised pictures of the Moon’s surface are beamed back to Earth in March 1965. The Ranger 9 uncrewed probe showed viewers unprecedented images of the lunar terrain.

SCIENCE AND ETHICS

Throughout history, the scientist’s quest for understanding has led many into serious trouble. In 1633, the Catholic Inquisition tried **Galileo** << 268–69 for heresy for suggesting that Earth moves around the Sun. **Charles Darwin’s theories of evolution** << 340–41 challenged “creationist” views that God created the world, triggering a huge debate << 342–43.

COMPUTERS

In 1801, **Joseph Jacquard** invented a power loom, which could base its weave on patterns read from punched wooden cards. **Charles Babbage** << 344 built on this concept in the 1830s, to create the **difference calculator**, which could store and calculate numbers. In 1890,



MARCONI

TRANSATLANTIC RADIO

In 1901, **Guglielmo Marconi** << 344–45, inventor of the radio-telegraph system, made the first transatlantic radio transmission, across a distance of 2,100 miles (3,400 km).

Uses for smart alloys are being investigated by the aircraft industry, seeking ways to improve aircraft maneuverability, and by the medical profession—for example, by incorporating flexibility into the metal plates and pins that are used to support broken limbs while they heal.

Progress in the understanding of natural polymers (compounds made up of simple repeating molecular units), led to the development of synthetic polymers, which are used in plastic-based

materials such as Plexiglas, polythene, and many modern textile fabrics. Commercial production of nylon, the first fiber to be made completely from petrochemicals (substances derived from petroleum), began in 1939. Nylon was the first of a new breed of manufactured

fibres, which include polyester, Lycra, and Kevlar. In 1986, microfibers further revolutionized fabric technology, producing materials made from minute filaments that are as

Apollo rocket

At 9:32 am on July 16, 1969 a plume of flame signals liftoff of the Apollo 11 Saturn V space vehicle. US astronauts Neil Armstrong, Michael Collins, and Buzz Aldrin leave the Kennedy Space Center, heading for the Moon.